



AIR ATLAS

An autonomous drone-based disaster response system that supports operators without delaying critical action

UI/UX CASE STUDY / OPERATOR CONSOLE

Problem Statement

In disaster scenarios such as wildfires, response time is critical but delays are rarely caused by slow detection or unskilled operators. Instead, time is lost to **decision fragmentation**: a series of small, sequential choices made under uncertainty, often across disconnected systems.

The operator is not overwhelmed by lack of information, but by the burden of having to assemble decisions manually while managing multiple zones, systems, and ongoing incidents.

A single alert initiates verification loops, permission checks, handoffs and escalations. By the time action is coordinated, the situation has already escalated.

PAIN POINTS IDENTIFIED

- 01 High cognitive load caused by fragmented decision-making
- 02 Uncertainty around AI confidence and system intent
- 03 Alert fatigue during multi-incident monitoring
- 04 Difficulty maintaining oversight of autonomous actions
- 05 Fear of making irreversible mistakes under time pressure

Our Solution

Decision compression,
not decision speed.

01 Pre-authorized response chains

Approved chains exist before the incident. The operator is not asked what to do, only whether to allow the response to continue.

02 A single irreversibility moment

One deliberate decision point showing what the system believes, what is already in motion, and what happens next if uninterrupted.

03 Everything is already moving

By the time the operator is notified, drones are airborne and feeds are streaming. Intervention is a conscious choice, not a prerequisite for action.

THE ONE QUESTION THE OPERATOR ANSWERS

INC-2026-0047 / WILDFIRE

● LIVE

Do I let this continue?

- ▶ 4 drones airborne · perimeter locked
- ▶ Ground team notified · ETA 6 min
- ▶ Evacuation trigger armed at T+90s

AI CONFIDENCE

94%

ALLOW

PAUSE

ABORT

SCOPE

Project Overview

Air Atlas compresses multiple high-stakes micro-decisions into one clear moment of human judgment, allowing operators to act decisively without sacrificing control or accountability.

ROLE UI/UX Designer — solo Research, IA, interaction, visual	DOMAIN Autonomous drone operations Wildfire disaster response	PLATFORM Desktop operator console Dark, control-room environment
METHOD Proxy & domain research Assumption-based reasoning	DELIVERABLES 9 hi-fidelity screens IA map, user flow, design system	TARGET Coordinated response in 90s Across multiple live incidents

DESIGN PHILOSOPHY

- Fewer decisions lead to faster action.** Reducing choice volume beats optimizing decision speed.
- Systems act first, humans judge.** Autonomy initiates; authority over escalation stays human.
- Clarity beats configurability.** In crisis, operators need certainty — not complex controls.

System Ideation

1 Pre-authorized response chains

Before incidents occur, Air Atlas operates on predefined and approved response chains — for example, wildfire near a residential area. Each chain includes:

- Autonomous drone dispatch
- Sensor and visual confirmation
- Escalation thresholds
- Communication and evacuation triggers

When an incident is detected, the system pre-commits to the appropriate chain.

2 Single irreversibility moment

Rather than presenting multiple reversible actions, the system creates one deliberate decision point. At this moment, the operator sees:

- What the system believes is happening
- What actions are already in motion
- What will happen next if uninterrupted

THE OPERATOR ANSWERS A SINGLE QUESTION

“Do I let this continue?”

System Ideation

3 Everything is already moving

By the time the operator is notified:

- ✓ Drones are airborne
- ✓ Feeds are streaming
- ✓ Risk projections are forming
- ✓ Escalation paths are prepared

The operator is never starting from zero. Intervention becomes a conscious choice, not a prerequisite for action.

4 Outcome-oriented experience

The interface prioritizes outcomes over mechanics:

- Projected disaster spread
- Time-to-impact windows
- Evacuation readiness
- Response progression timelines

Technical details remain accessible but secondary, keeping the operator focused on consequences rather than execution.

5 AI as a decision editor

Not a decision replacement:

- AI assembles the first draft
- Humans approve, pause or abort
- Reasoning stays auditable
- Intervention drops as trust grows

This human-in-the-loop model ensures safety, adaptability and accountability in life-critical operations.

Research Summary

The objective was to understand how emergency operators make critical decisions under extreme time pressure, how they interact with autonomous systems, and what slows response despite advanced sensing — then find ways to reduce response time without increasing cognitive overload, false confidence, or loss of human control.

1 Domain & proxy research

Established research, articles, workflows and interfaces from comparable environments where rapid decision-making and human–AI collaboration are essential:

Emergency dispatch systems (911, 112)

Air traffic control aviation monitoring

Military and disaster response coordination

Autonomous vehicle supervision and autopilot

2 Assumption-based reasoning

Where direct user access was not possible, assumptions were drawn from consistent patterns across emergency response systems:

Operators manage multiple zones simultaneously

False positives are costly and create hesitation

Clarity is valued over raw data volume

Automation helps only when it is interruptible

What the research showed

FINDING 01

Operators are required to make multiple small decisions in rapid succession during emergencies.

INSIGHT

Decision paralysis is caused more by the number of decisions than by their complexity.

DESIGN IMPLICATION

Compress micro-decisions into a single, clearly framed moment of human judgment.

FINDING 02

Manual verification loops significantly delay response even when sensor confidence is high.

INSIGHT

Verification is driven by fear of being wrong, not by lack of data.

DESIGN IMPLICATION

Let autonomy act immediately while surfacing confidence and consequences rather than raw sensor data.

FINDING 03

Operators struggle to track what autonomous systems are already doing during active incidents.

INSIGHT

Loss of situational clarity leads to hesitation and duplicate actions.

DESIGN IMPLICATION

Present all autonomous actions as a single coordinated response, not separate system activities.

What the research showed

FINDING 04

Operators think in terms of outcomes and risk, not technical execution.

INSIGHT

They care more about “what happens next” than “how it happens.”

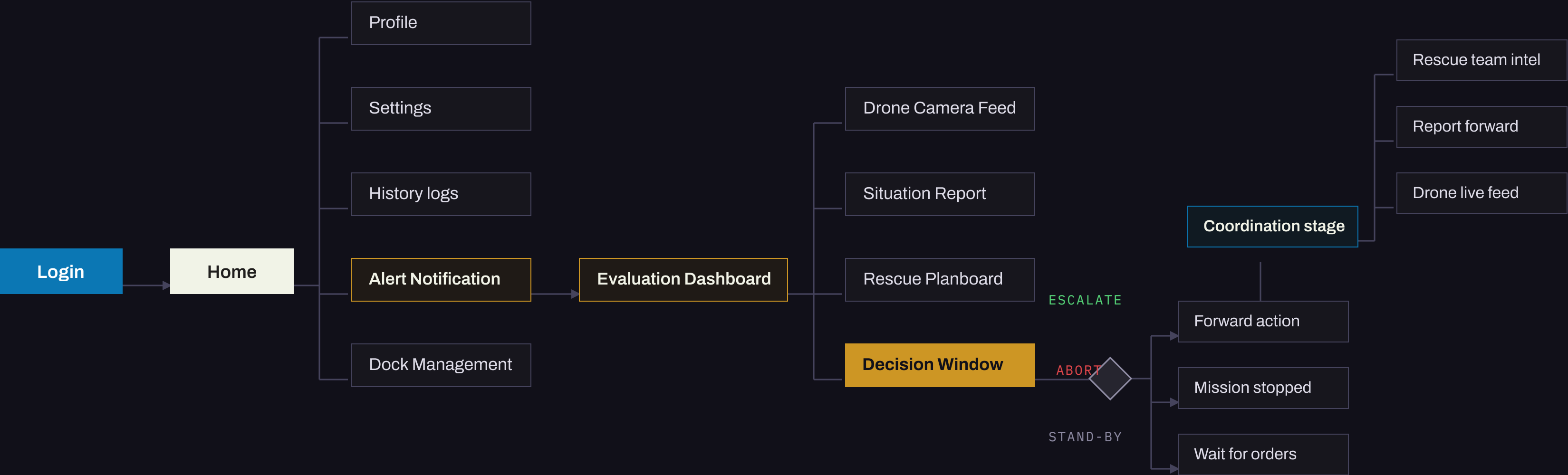
DESIGN IMPLICATION

Shift from control-based interactions to outcome-based summaries and future projections.

OPPORTUNITY AREAS

- Decision compression through pre-authorized response chains
- AI-generated action summaries instead of step-by-step controls
- Single incident takeover views during critical alerts
- Timeline-based presentation of “what happens next”
- Clear human override points instead of continuous micromanagement

Information Architecture and User Flow



Getting in, getting oriented

Sign-in asks for an operator ID and a code, nothing more. Home opens straight onto the patrol map — the fleet is already flying, so there is nothing to start.

The image is a composite of two digital interfaces. The left interface is a login screen for 'flytbase'. It features a dark blue header with the 'flytbase' logo. Below the header, the word 'OPERATOR' is displayed in a light blue, monospace font. A large, white 'Login' text is centered. Underneath, there is a form with two fields: 'Operator ID' containing the text 'OP-4471' and a 'Login code' field with four input boxes, the last of which contains a vertical bar. A prominent orange 'Login' button is at the bottom. The footer shows 'Control room · Sector 4' and 'Shift 14:00-22:00'. The right interface is a 'City patrolling' dashboard. It has a dark blue sidebar with a menu including 'Flight Center', 'Fleet' (highlighted in orange), 'Planning', 'Device Management', 'Zones', 'Logs', 'Integrations', 'Support', and 'Settings'. The main area displays an aerial map of a residential area with three drone locations marked by colored squares and labeled: 'DRN-415' (pink), 'DRN-432' (yellow), and 'DRN-441' (cyan). A green box at the top of the map indicates '4 DRONES PATROLLING'. A search bar at the top right says 'Search location'. At the bottom, a status bar shows 'AIRBORNE 04', 'DOCKED 12', 'ANOMALIES 00', and 'WIND 11 KMPH'. The user's name 'Olivia Rhye' and email 'olivia@flytbase.com' are in the bottom left of the sidebar.

The alert takes over

The notification opens with actions already taken, not with a request for permission. The decision window then reduces the whole response to one irreversible choice.

INCIDENT ID

INC-2026-0047

CATEGORY

WILDFIRE

STATUS

ACTIVE RESPONSE

TIME ELAPSED

00:00:18

SEVERITY

▲ HIGH

RESPONSE STAGE

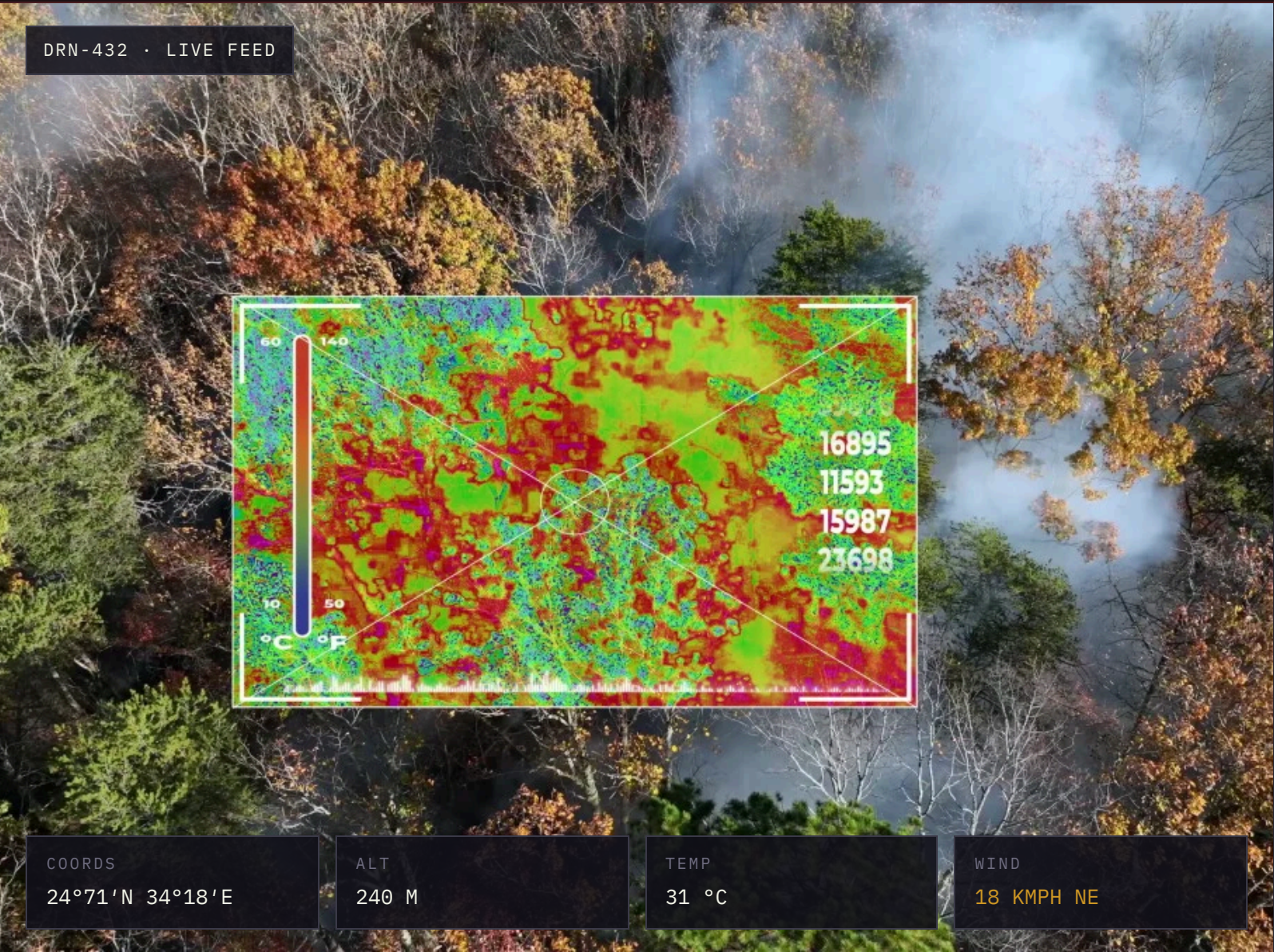
ALERT STAGE

Anomaly detected · Wildfire, Zone 04

1.2 KM FROM RESIDENTIAL PERIMETER · RESPONSE CHAIN ALREADY EXECUTING

MUTE SIREN

DRN-432 · LIVE FEED



COORDS

24°71'N 34°18'E

ALT

240 M

TEMP

31 °C

WIND

18 KMPH NE

ALREADY IN MOTION

T+00

Anomaly validated by AI · 96%

T+04

4 drones launched autonomously

T+11

Thermal perimeter established

T+18

Ground team notified · ETA 6 min

T+90

Evacuation trigger armed

IF UNINTERRUPTED

Spread reaches the residential perimeter in **14 minutes**. Evacuation alarm fires automatically.

AI CONFIDENCE

96%

REVIEW INCIDENT

DECISION WINDOW · THE SINGLE MOMENT

INC-2026-0047

● AWAITING JUDGMENT

Do I let this continue?

Confirmed wildfire spreading NE toward Zone 04 housing. The response chain will continue on its own unless you intervene.

CONSEQUENCES OF CONTINUATION

▶ Evacuation alarm to 340 residents

▶ 2 additional docks released

▶ Fire service dispatch confirmed

AUTO-CONTINUE IN

00:21

ALLOW · CONTINUE

PAUSE

ABORT

Analyze, then coordinate

Outcomes lead: spread projection, time-to-impact, evacuation readiness. Raw telemetry stays one glance away. The report is assembled from what the operator confirmed.

INCIDENT ID

INC-2026-0047

CATEGORY

WILDFIRE

STATUS

ACTIVE RESPONSE

TIME ELAPSED

00:03:11

SEVERITY

▲ HIGH

RESPONSE STAGE

ANALYZING STAGE

RGB VIEW

MULTI VIEW

FULL SCREEN

THREAT DETECTION ▼

SPREAD PROJECTION

16895

11593

15987

23698

DRN-432

ZONE 04 · PERIMETER

240 M

AVAILABLE

12

UNITS DOCKED

TIME TO IMPACT

14 MIN

TO RESIDENTIAL EDGE

TEMP

31 °C

COORDS

24°71'N

HEAT SIGNAL

STRONG

EVAC READY

44 %

340 RESIDENTS

Fire started : 3 min ago

Status : Spreading rapidly

Pact : Call support

SELECT ACTION

COORDINATION · SITUATION REPORT

• SHARED WITH GROUND TEAM

Detection : Fire

Situation Report

AUTO-COMPILED

SENT TO

Ground Team Alpha · Fire Service 04

CONFIRMED

Wildfire, 96% · OP-4471 at 00:03:24

ATTACHED

Thermal map, spread projection, route

CONTROL ROOM CHANNEL

GROUND ALPHA · 00:04

Received. Routing via service road, ETA 6 min.

SYSTEM · 00:05

Perimeter drones repositioned to NE flank.

FIRE SERVICE 04 · 00:06

Two units dispatched. Standing by for evac call.

SOUND EVAC ALARM

FORWARD

The quiet state

Most of the time nothing is burning. Docks, devices and history use the same grid and the same telemetry language, so the alert state is a change in urgency, not a change in system.

DOCK MANAGEMENT

DRN-432

DRN-415

DRN-427

DRN-438

DRN-441

DRN-443

DRN-465

DRN-447

CHARGE STATUS

READY

10

CHARGING

02

0P-4471 · 14:02

DRN-438 · MAINTENANCE DUE

DEVICE MANAGEMENT

flytbase

Flight Center

Planning

Devices

Sensors

Sites

Map models

Zones

Logs

Integrations

Manage Device

+ Add new device

View and manage all your devices

Drone and Dock

Payload

All

Search for devices

ALL DEVICES

DEVICE	NAME	MODEL	FIRMWARE	
DRN-432	Zone 04	Matrice 30T	v12.09.08	View
DOCK-02	Zone 04	Dock 2	v12.09.08	View
DRN-415	Zone 02	Matrice 30T	v12.09.08	View
DRN-438	Maintenance	Matrice 30T	v11.42.01	View
SNR-M02	Thermal array	M2E Sensor	v08.11.00	View

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HISTORY LOGS

 flytbase Flight Center Fleet Planning Devices Logs Zones Integrations	History logs Every decision recorded against an operator ALL ALLOWED ABORTED AUTO <table> <tr> <th>INCIDENT</th><th>ELAPSED</th><th>OUTCOME</th></tr> <tr> <td> INC-2026-0047 Wildfire · Zone 04 </td><td>00:03:24</td><td> ALLOWED </td></tr> <tr> <td> INC-2026-0046 Smoke signature · Zone 02 </td><td>00:01:02</td><td> AUTO-RESOLVED </td></tr> <tr> <td> INC-2026-0045 False positive · Zone 07 </td><td>00:00:48</td><td> ABORTED </td></tr> <tr> <td> INC-2026-0044 Wildfire · Sector 11 </td><td>00:04:10</td><td> ALLOWED </td></tr> <tr> <td> INC-2026-0043 Structure heat · Zone 01 </td><td>00:02:19</td><td> AUTO-RESOLVED </td></tr> </table> LAST 30 DAYS 47 39 04 INCIDENTS ALLOWED ABORTED	INCIDENT	ELAPSED	OUTCOME	INC-2026-0047 Wildfire · Zone 04	00:03:24	ALLOWED	INC-2026-0046 Smoke signature · Zone 02	00:01:02	AUTO-RESOLVED	INC-2026-0045 False positive · Zone 07	00:00:48	ABORTED	INC-2026-0044 Wildfire · Sector 11	00:04:10	ALLOWED	INC-2026-0043 Structure heat · Zone 01	00:02:19	AUTO-RESOLVED
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OUTCOME

Why this direction works

- 01 Collapses response time by eliminating decision fragmentation
- 02 Reduces cognitive load without reducing human authority
- 03 Enables rapid, coordinated drone response in the first 90 seconds
- 04 Scales across multiple incidents and monitoring zones
- 05 Aligns with FlytBase's positioning as a reliable, enterprise-grade autonomous operations platform



Thank you

Air Atlas — an autonomous drone-based disaster response system designed around one moment of human judgment.

KSHITIZ KUMAR TIRKEY